WASP-11b

R-0766

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_231025` · created 2026-07-28T04:33:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460242.8945 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.052
Transit depth (Rp/Rs)^2	3.07 +/- 1.82 [%]
Orbital Inclination (inc)	89.74 +/- 1.63
Airmass coefficient 1 (a1)	0.42 +/- 0.22
Airmass coefficient 2 (a2)	0.72 +/- 0.52
Scatter in the residuals of the lightcurve fit is	54.91 %
Best Comparison Star	None
Optimal Aperture	5.26
Optimal Annulus	8.66
Transit Duration (day)	0.106 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.052 vs 0.131 (deviation 0.43σ)
Duration fit vs published	0.106 d vs 0.1075 d (deviation 0.07σ)
Mid-time O–C	-6.8 min
Depth SNR	1.7
Residual scatter	54.91 %

Light curve

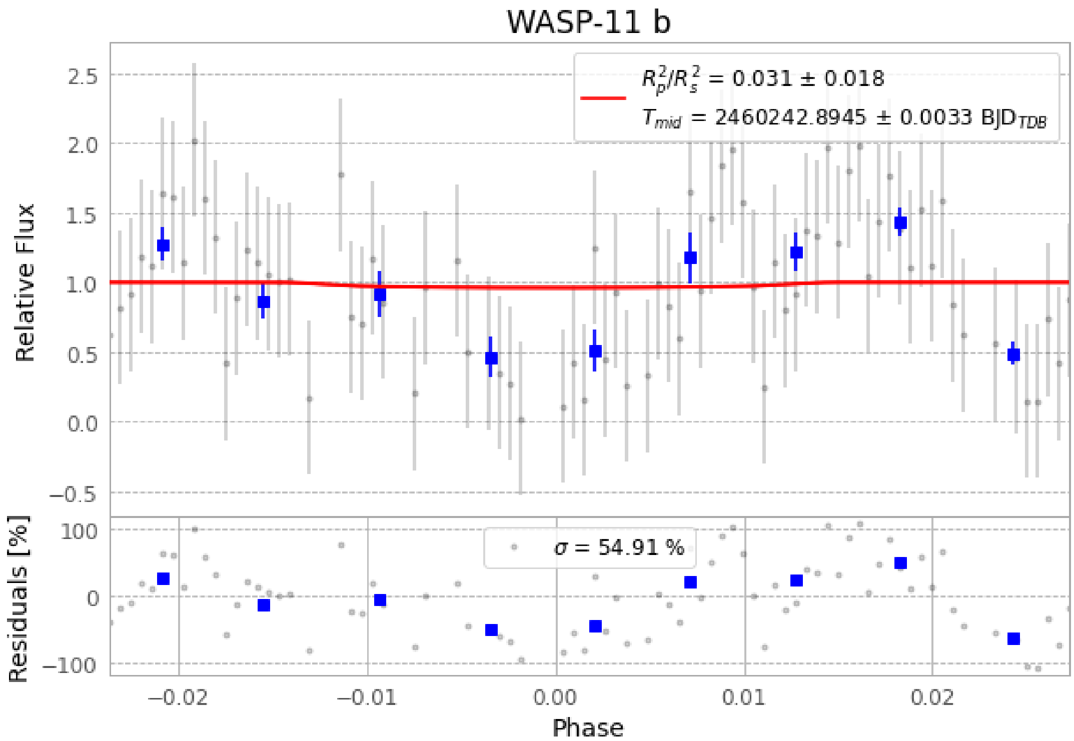

Figure 1 — FinalLightCurve_WASP-11 b_2023-10-25.png (SHA-256 7cd9df3ffb46631d…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [604, 406], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 fc5797adf28254d7…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

92 FITS frames (60 MB), `HATP-10231025071216.FITS` → `HATP-10231025114515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-231025.log (141d0037d378…), FinalLightCurve_WASP-11 b_2023-10-25.png (7cd9df3ffb46…), FinalLightCurve_WASP-11 b_2023-10-25.csv (c79ddbb8d488…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)